

CUSTOMER ANALYSIS PROJECT

Python(pandas) , SQL , Power bi

Customer Analysis

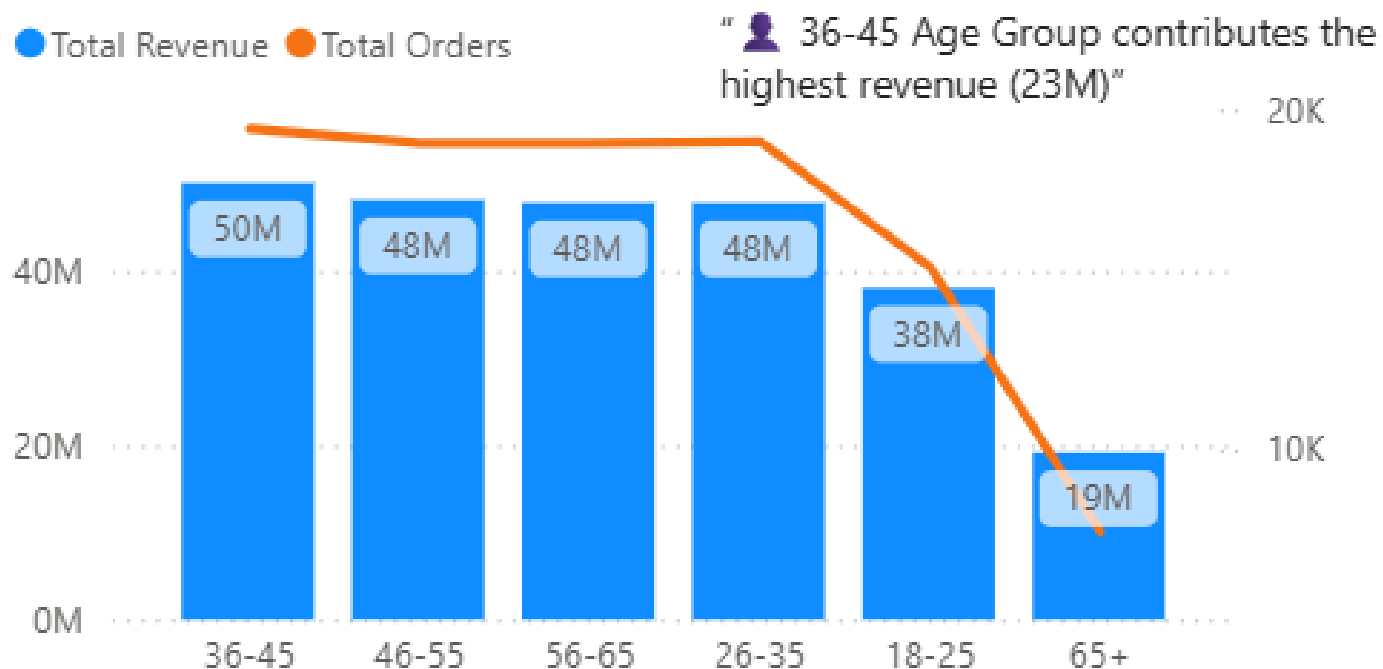
Revenue
251.5M

Quantity Sold
299K

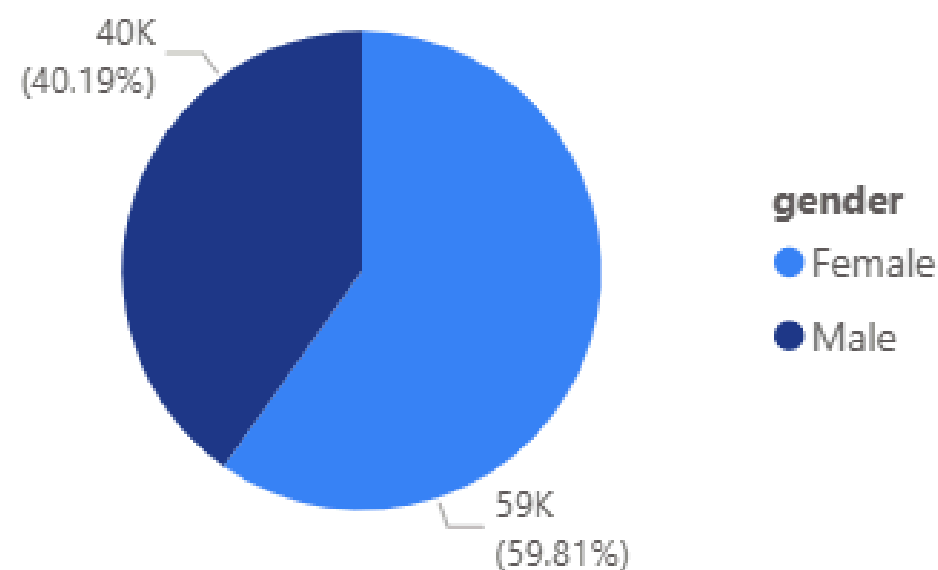
Customers
99K

Avg_Spend per cust.
2.53K

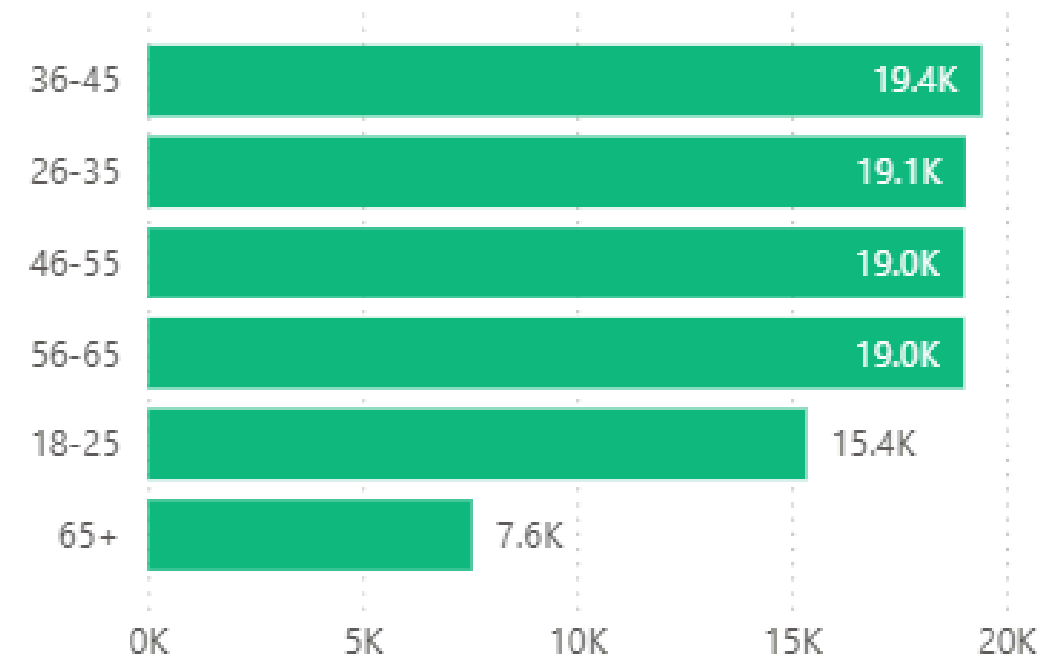
Total Revenue & Orders by age_groups



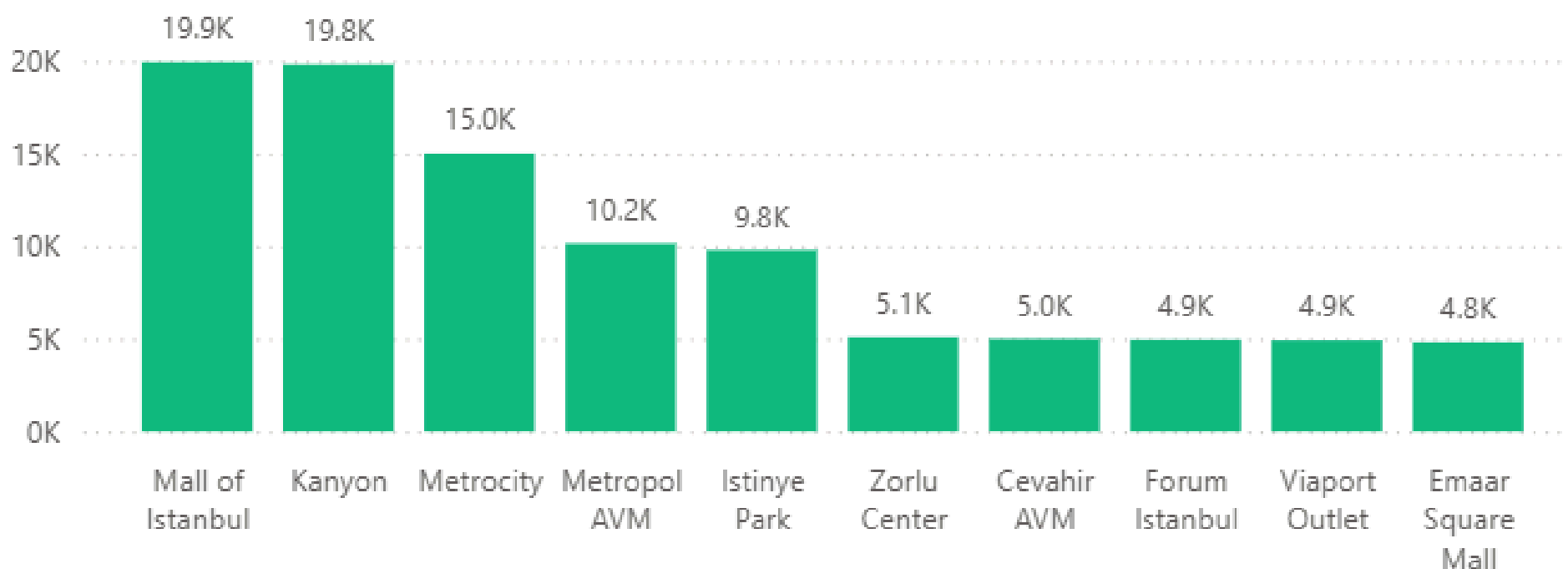
Total Customers by gender



Total Customers by age_groups



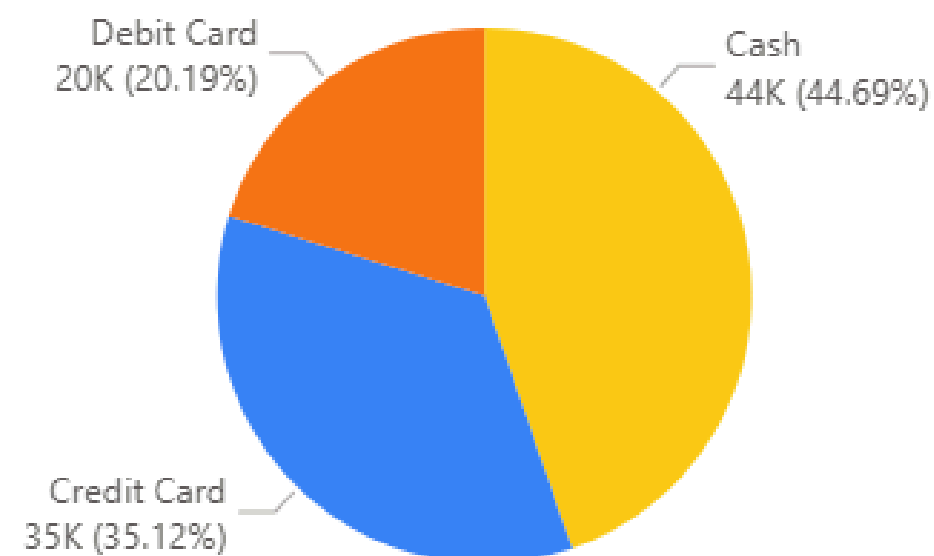
Total Customers by shopping_mall



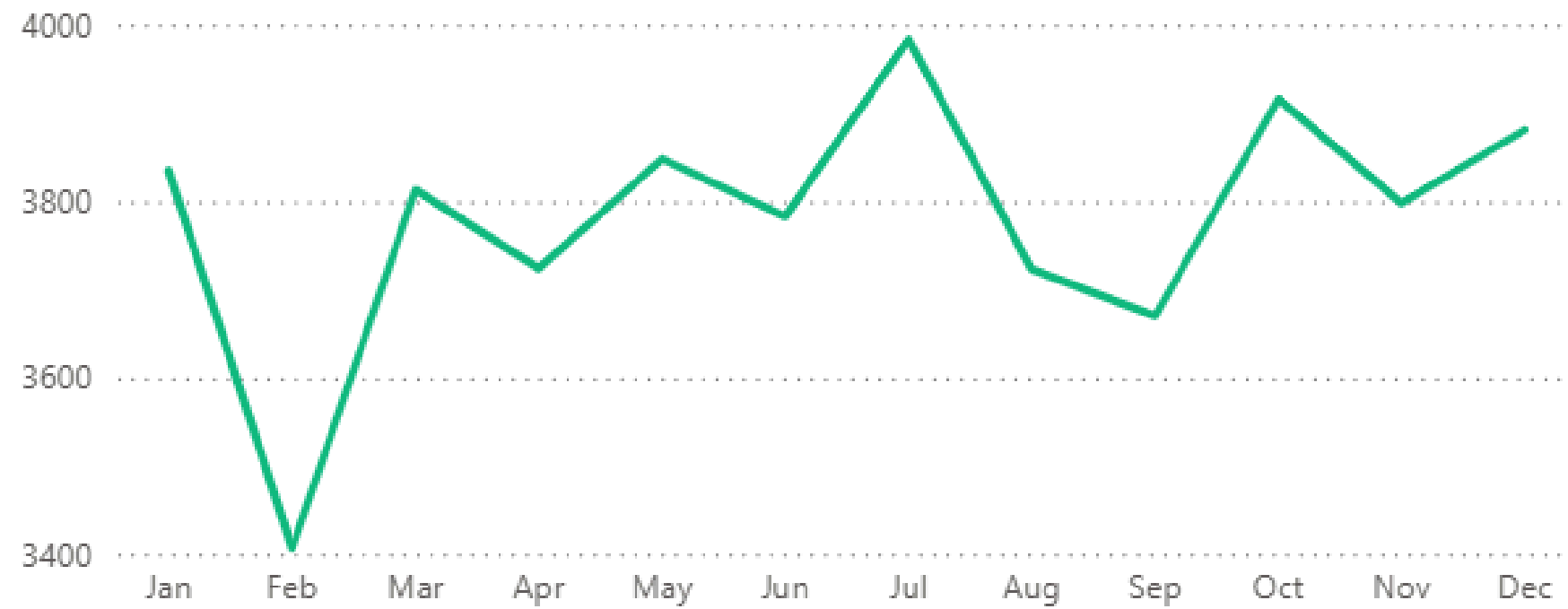
category

- ☐ Books
- ☐ Clothing
- ☐ Cosmetics
- ☐ Food & Beverage
- ☐ Shoes
- ☐ Souvenir
- ☐ Technology
- ☐ Toys

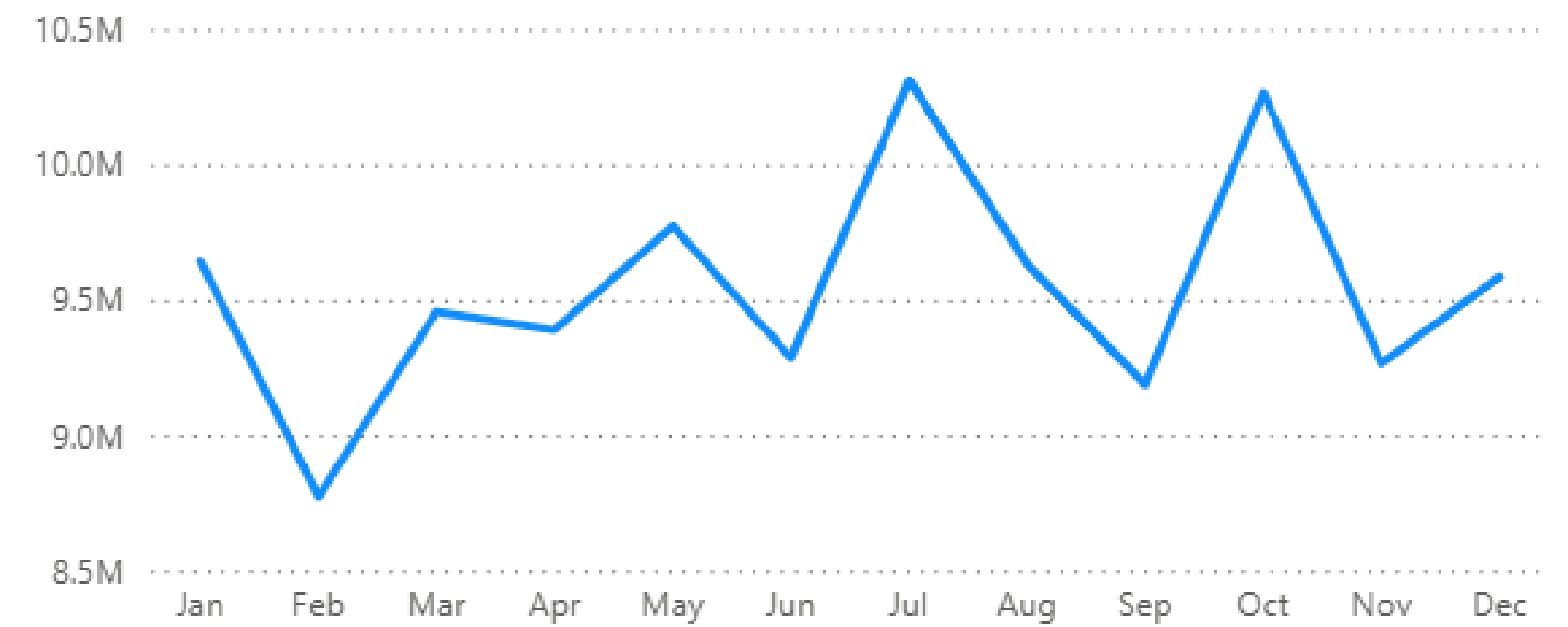
payment_method used by Customers



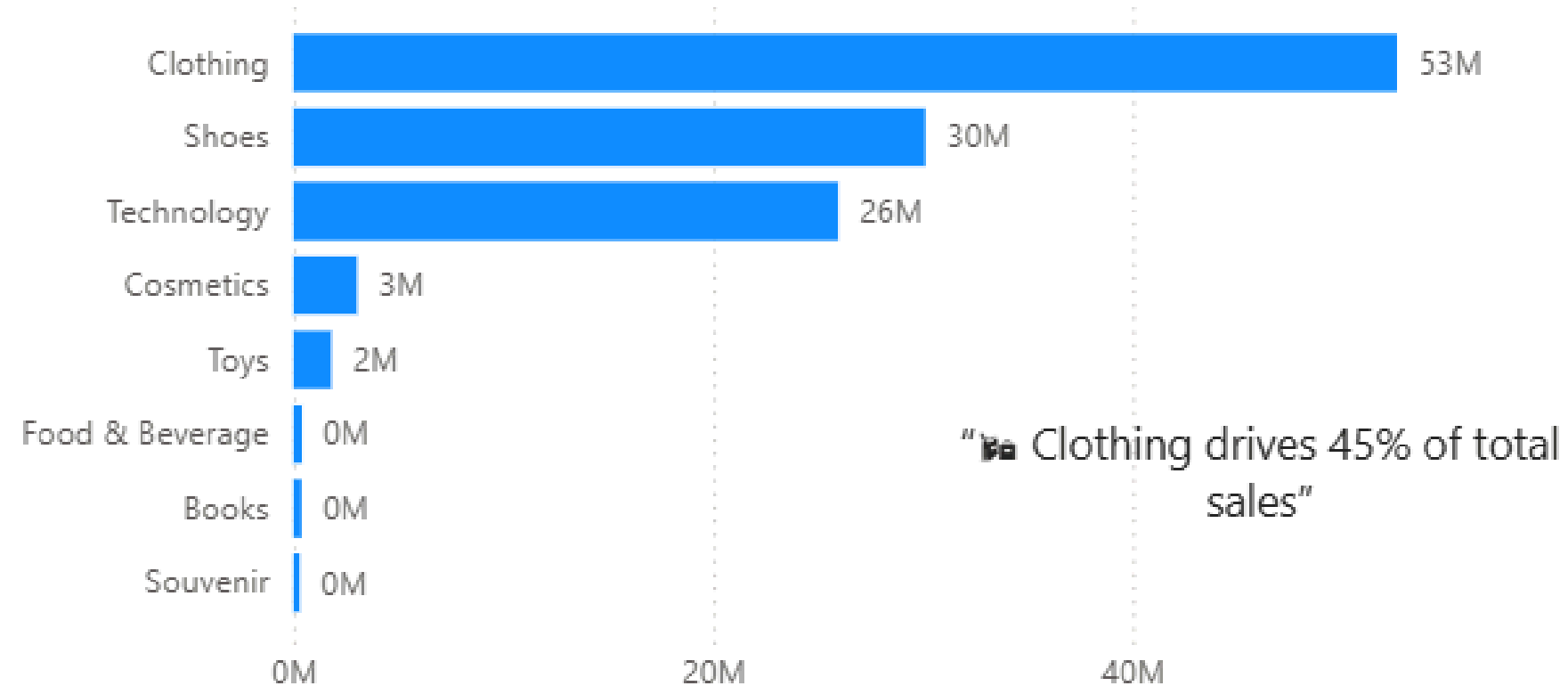
Total Customers by Month



Total Revenue by Month



Total Revenue by category



year

2021

2022

2023

Month Name	Total Revenue	Total_quantity sold
Jan	9641614.6	11483
Feb	8772315.2	10179
Mar	9455359.4	11383
Apr	9389541.5	11270
May	9771757.0	11620
Jun	9286271.4	11352
Jul	10311119.7	11986
Aug	9630655.7	11205
Sep	9188165.6	10901
Oct	10263015.1	11711
Nov	9265555.3	11355
Dec	9585200.2	11651

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```
1  import pandas as pd
2
3  customer = pd.read_csv('customer_shopping_data.csv')
4
5  # Convert invoice_date to datetime
6
7  customer['invoice_date'] = pd.to_datetime(customer['invoice_date'], format='%d/%m/%Y' ,errors='coerce')
8
9
10 # Remove duplicates (same invoice, customer, category, date)
11
12 customer = customer.drop_duplicates(subset=('invoice_no', 'customer_id', 'category', 'invoice_date'))
13
14 # Create calculated columns
15
16 customer['total_spend'] = customer['quantity']*customer['price']
17
18 customer['year'] = customer['invoice_date'].dt.year
19 customer['month'] = customer['invoice_date'].dt.month
20 customer['year_month'] = customer['invoice_date'].dt.to_period('M')
21
22 |
23 # Create Age Groups
24
25 bins = [0, 17 , 25 ,35, 45, 55, 65 ,100]
26 labels= ['<18' , '18-25' , '26-35', '36-45' , '46-55' , '56-65' , '65+' ]
27
28 customer['age_groups'] = pd.cut(customer['age'], bins=bins, labels=labels, right=True)
29
30 print(customer.head())
31
32 # Save cleaned data to CSV
33
34 customer.to_csv("customer_data_cleaned.csv", index=False)
35
36 print("Saved")
```

```
1 • create database customer_analysis;
2 • use customer_analysis;
3
4 • select * from customer;
5
6   -- Total Customers --
7
8 • SELECT
9     COUNT(DISTINCT customer_id) AS total_customer
10  FROM
11     customer;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

total_customer
99457


```
13      -- Gender Distribution --
14
15 •   SELECT
16      gender, COUNT(DISTINCT customer_id) AS total_customer
17 FROM
18      customer
19 GROUP BY gender;
```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	gender	total_customer
	Female	59482
	Male	39975

```

21  -- Revenue by Age Group --
22
23  •  SELECT
24      age_groups, ROUND(SUM(total_spend), 2) AS total_revenue
25  FROM
26      customer
27  GROUP BY 1
28  ORDER BY 2 DESC;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	age_groups	total_revenue
▶	36-45	50184235.79
	46-55	48219742.64
	56-65	47893010.77
	26-35	47879659.58
	18-25	38118271.35
	65+	19210874.12

```

30      -- Top 5 Spending Customers --
31
32 •    SELECT
33         customer_id, ROUND(SUM(total_spend), 2) AS total_spend
34     FROM
35         customer
36     GROUP BY 1
37     ORDER BY 2 DESC

```

Result Grid |  Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows:

	customer_id	total_spend
▶	C326173	26250
	C101667	26250
	C163850	26250
	C150501	26250
	C106836	26250


```

40  -- Revenue by Category --
41
42  •  SELECT
43      category, ROUND(SUM(total_spend), 2) AS total_revenue
44  FROM
45      customer
46  GROUP BY 1
47  ORDER BY 2 DESC;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	category	total_revenue
•	Clothing	113996791.04
	Shoes	66553451.47
	Technology	57862350
	Cosmetics	6792862.9
	Toys	3980426.24
	Food & Beverage	849535.05
	Books	834552.9
	Souvenir	635824.65

```

49  -- Monthly Revenue Trend --
50
51  •  SELECT
52      month, ROUND(SUM(total_spend), 2) AS total_revenue
53  FROM
54      customer
55  GROUP BY 1
56  ORDER BY 2 DESC;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	month	total_revenue
	1	28891525.59
	2	26625090.1
	3	21956191.33
	10	20545090.43
	7	20378722.63
	5	19719331.1
	12	19455085.64
	8	19282361.29
	6	18933775.3
	9	18795794.91

```

58      -- Most Popular Payment Method --
59
60 •   SELECT
61         payment_method, COUNT(*) AS total_transaction
62     FROM
63         customer
64     GROUP BY 1
65     ORDER BY 2 DESC;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content: 

	payment_method	total_transaction
	Cash	44447
	Credit Card	34931
	Debit Card	20079

```

67      -- Top 5 Shopping Malls by Revenue--
68
69 •   SELECT
70         shopping_mall, ROUND(SUM(total_spend), 2) AS total_revenue
71     FROM
72         customer
73     GROUP BY 1
74     ORDER BY 2 DESC
75     LIMIT 10;

```

Result Grid |   Filter Rows: | Export:  | Wrap Cell Content:  | Fetch rows:

	shopping_mall	total_revenue
	Mall of Istanbul	50872481.68
	Kanyon	50554231.1
	Metrocity	37302787.33
	Metropol AVM	25379913.19
	Istinye Park	24618827.68
	Zorlu Center	12901053.82
	Cevahir AVM	12645138.2
	Viaport Outlet	12521339.72
	Emaar Square Mall	12406100.29
	Forum Istanbul	12303921.24

**Thank you
very much!**